

an operator or identifier used in a Dart program.

Another interesting feature of Dart is 'Method Cascading'. To perform this, the operator used is `'..'`. It is the double dot operator.

An overview of the Dart programming constructs is provided at <https://www.dartlang.org/guides/language/language-tour>.

As stated in the beginning of this article, Dart is a general-purpose programming language. It is used to build applications belonging to various types. The three major types of apps that can be built with it are listed below:

- Dart Web apps
- Dart Mobile apps
- Dart Server (or command line) apps

Dart Web apps

As per the official documentation (<https://webdev.dartlang.org/>), Google uses Dart for building critical Web apps. It is utilised with AngularDart. Google's internal CRM is also built using Dart. Apart from the search giant, many other reputed companies also use Dart for their critical Web apps.

In the case of Dart Web apps, the contents of the HTML elements can be manipulated dynamically from Dart code. A simple 'countdown' timer code follows.

In HTML, type:

```
<h1 id="countdown"></h1>
```

In Dart, type:

```
import 'dart:html';

main() async {
  var countdown = querySelector("#countdown");
  for (int i = 100; i >= 0; i--) {
    countdown.text = "Time: $i";
    await window.animationFrame;
  }
}
```

Place this code in DartPad and click *Run*. In the output window (HTML), you will notice the countdown from 100 to 0. The HTML element `'<h1>'` is accessed with the `querySelector()` function. Here, the `'id'` is supplied as input. Then the contents are dynamically modified by using the `'text'` attribute.

The `dart:html` library is utilised in the above program. It is a wrapper around the DOM (Document Object Model) and Window API. Due to the availability of this wrapper library, the developer need not worry much about browser support.

Important advantages of building Web apps with Dart are:

- The Dart SDK is of production quality. It is stable and maintained.
- With Angular, Dart becomes a powerful combination.

Detailed information regarding building Web apps with Dart

is provided at <https://webdev.dartlang.org/guides/get-started>.

Dart mobile apps

Dart mobile apps are built using Flutter, which is a SDK for building mobile apps. For iOS and Android, the apps can be built with a single codebase. Apps built using Flutter are high-performance and high-fidelity. The major advantages of building mobile apps with Flutter are as follows:

- iOS and Android apps can be built using the same codebase.
- Prototyping is comparatively simpler.
- The apps are built with highly customised user experiences. The material design widgets for the Flutter framework are handy. The custom designs can also be implemented using OEM widget sets. Flutter includes the following components:
 - A functional reactive framework
 - A 2D rendering engine
 - Ready-to-use widgets
 - Development tools

In the application development model of Flutter, everything is a widget. Flutter has a unified object model in which all components are widgets. In Flutter terminology, a widget can be used to define all of the following: structural elements, style elements, layout, and even the business logic.

Fundamental Dart libraries

The three most fundamental Dart libraries are listed below:

- `dart:core` - All basic functionalities of Dart such as strings, collections, dates and URIs are processed using this library.
- `dart:html` - This library is the wrapper for DOM manipulation which is much used with Web apps.
- `dart:io` - This library is used with command line apps.

The libraries in Dart are imported with the help of `'import'`, as shown below:

```
import 'dart:html';
import 'dart:math';
```

Apart from the above-mentioned core libraries, there are various other libraries. Some of them are listed below:

- `dart:async` - This is a library that supports asynchronous programming.
- `dart:developer` - This library is used for interacting with developer tools such as the debugger and inspector.
- `dart:js` - This library provides support for interoperating with JavaScript.
- `dart:svg` - This is used to handle support vector graphics.
- `dart:web_audio` - This is used to perform audio programming in the browser.
- `dart:web_gl` - This is to perform 3D programming in the browser.